Course Summary - MATH 259

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Abstract

These are the most important definitions and results from this iteration of Honors Algebra III (Spring 2019) at The University of Chicago. Any merits should be attributed to the lecturer, and any mistakes should be attributed to the notetaker.

Basic Field Extensions

Definition 1.

- E is a *field extension* of F if E is a field and F is a subfield of F; this is usually written E/F.
- E/F is *finite* if $\deg(E/F) := \dim_F(E)$ is finite.
- E/F is *algebraic* if each $x \in E$ is the root of some $f \in F[t]$. $x \in E$ is algebraic if it is the root of some $f \in F[t]$.

Lemma 2. *E/F is algebraic iff for each $\alpha \in E$, the collection $\{\alpha^n | n \in \mathbb{N}_0\}$ spans a finite dimensional F-vector space.*

Definition 3. Let $\alpha \in E/F$ be algebraic. g is the *minimal polynomial of* α if g is monic, $g(\alpha) = 0$, and $g|f$, for any $f \in F[x]$ such that $f(\alpha) = 0$. Such a g is usually denoted $\mathrm{mp}_F(\alpha)$.

Proposition 4 (Tower Theorem)**.** *$F \subseteq E \subseteq K$ as fields, K/E, E/F finite. Then, K/F is a finite extension and*

$$\deg(K/F) = \deg(K/E)\deg(E/F). \tag{1}$$

Lemma 5. *Finite extensions are algebraic.*

Theorem 6. *E/F, $E^C = \{\alpha \in E | \, \alpha$ algebraic over $F\}$. Then, E^C is a subfield of E.*

Definition 7. Let $\alpha \in E/F$, $g = \mathrm{mp}_F(\alpha)$. Then, $F(\alpha) = E/(g)$ is the smallest subfield of E containing α.

Definition 8. F field, f monic $\in F[x]$, $\deg(f) \geq 1$. E/F is a splitting field of f if

$$f = \prod_{i=1}^{d} (x - \alpha_i) \in E[x] \tag{2}$$

and $E = F(\alpha_1, \ldots, \alpha_d)$. We usually write $E = \mathrm{Spl}_F(f)$.

Theorem 9. *Splitting fields exist and are unique up to isomorphism that fixes the base field.*

Theorem 10. *$\#\mu_n(F) = n$, $0 \neq a \in F$, $E = F(\alpha^{1/n})$. Then, $\deg(E/F) = \deg(\mathrm{mp}_F(\alpha)) = d|n$, and there exists $b \in F$ such that $b^{n/d} = a$. If $h \in F$ has a dth root in E, then $h = c^d b^k$ for some $c \in F, k \in \mathbb{N}$.*

Theorem 11. *p prime, $k \in \mathbb{N}$, $q = p^k$, $E = \mathrm{Spl}_{\mathbb{F}_p}(x^q - x)$. E is a finite field with q elements. Furthermore, there exists a unique field with q elements.*

Theorem 12 (Lagrange's Theorem for Cyclic Extensions)**.** *F a field, $n \in \mathbb{N}$, $\#\mu_n(F) = n$. $F_0 \subseteq E$, $\sigma \in \mathsf{Aut}_F(E)$. Assume $\sigma^{\circ n} = \mathsf{id}_E$, but $\sigma^{\circ k} \neq \mathsf{id}_E$ for any $0 \leq k \leq n$. Let $F = \{b \in E | \sigma(b) = b\}$. Then, E is a field extension of F with degree n, and there exists $E = F(\alpha)$ with $\alpha^n = a \in F$.*

Theorem 13 (Converse)**.** $\mathrm{char}(E) = p > 0$, *$\sigma \in \mathsf{Aut}_F(E)$ with order n. Let $F = \{c \in E | \sigma(c) = c\}$. Then, $E = F(\alpha)$, $\alpha^p - \alpha = a \in F$.*

Definition 14. A field F is *perfect* if $\mathrm{char}(F) = 0$ or $\mathrm{char}(F) = p > 0$ and Frob_p is a surjection.

Lemma 15. *E/F finite extension, F perfect. Then, E is perfect.*

Theorem 16 (Primitive Element Theorem)**.** *Let E/F be a finite, separable extension. Then, there exists $\alpha \in E$ such that $E = F(\alpha)$.*

Galois Theory

Definition 17. E/F is *normal* if $\sigma(E) = E$ for all $\sigma \in \mathsf{Aut}_F(E)$.

Theorem 18. *A finite extension E/F is normal iff $E = \mathrm{Spl}(f)$ for some $f \in F[x]$.*

Definition 19. E/F field extension, $\alpha \in E$.

1. $\alpha \in E$ is *separable* over F if α is algebraic over F and $\mathrm{mp}_F(\alpha)'$ is nonzero evaluated at α.

2. E is *separable* if each $\alpha \in E$ is separable over F.

Definition 20. A finite extension E/F is *Galois* if it is normal and separable. If E/F is Galois, we define $\mathsf{Gal}(E/F) := \mathsf{Aut}_F(E)$.

Proposition 21.

1. $\#\mathsf{Gal}(E/F) = \deg(E/F)$.

2. $F \subseteq E \subseteq K$ *as subfields,* K/E *Galois. Then,* K/E *is Galois.*

Definition 22. Let $G \subseteq \mathsf{Aut}_E(K)$. Then, the *fixed field* K^G is $K^G = \{a \in K | \sigma(a) = a \text{ for all } \sigma \in G\}$.

Proposition 23. *If* K/F *is finite Galois with* $G = \mathsf{Gal}(K/F)$*, then* $K^G = F$.

Theorem 24 (Fundamental Theorem of Galois Theory)**.** *Let* K/F *be a finite Galois extension with* $G = \mathsf{Gal}(K/F)$*. Then, there is a direction-reversing bijection*

$$\{\textit{subgroups } \Gamma \leq G\} \leftrightarrow \{\textit{subfields } E \,|\, F \leq E \leq K\} \tag{3}$$

given by $E \mapsto \mathsf{Gal}(K/E)$, $\Gamma \mapsto K^\Gamma$.

Theorem 25 (E. Artin)**.** *Let* G *be a finite group of automorphisms acting faithfully on a field* K*, and let* $F = K^G$*. Then,* K *is a finite Galois extension of* F *and the embedding* $G \hookrightarrow \mathsf{Gal}(K/F)$ *is onto.*

Furthermore, if K *is a Galois extension of* F *and* $H \leq \mathsf{Gal}(K/F)$*, then* $\mathsf{Gal}(K/K^H) = H$.

Hilbert Theorem 90

Definition 26. Let $R \subseteq S$ be a subring of a commutative ring. Suppose that S is a free R-module of rank n, i.e. $S \xrightarrow{\phi} R^n$. Given $\alpha \in S$, consider the map $l_\alpha(x) = \alpha x$ for all $x \in S$. This pulls back to a map $\phi^{-1} l_\alpha \phi : R^n \to R^n$. Define

1. $\mathsf{tr}_R^S(\alpha) := \mathrm{tr}(\phi^{-1} l_\alpha \phi)$

2. $\mathsf{Norm}_R^S(\alpha) := \det(\phi^{-1} l_\alpha \phi)$.

If $\alpha \in R$, $\mathsf{Norm}_R^S(\alpha) = \alpha^n$.

Lemma 27. $\mathrm{char}(F) = 0, K = \mathrm{Spl}_F(x^n - 1)$. *Then,* $\mathsf{Gal}(K/F) \hookrightarrow (\mathbb{Z}/n\mathbb{Z})^\times$.

Theorem 28. $F = \mathbb{Q}$, $\mathsf{Gal}(K/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$.

Theorem 29 (Dedekind's Linear Independence of Characters)**.** *Let H be a group, K be a field. Let $\chi_1, \ldots, \chi_n : H \to K^\times$ be group homomorphisms. Then, $\chi_1, \ldots, \chi_n$ are linearly independent as functions $H \to K^\times$.*

Proposition 30. *Let $G = \{\sigma_1, \ldots, \sigma_n\}$. Then, every F-linear transformation $T : K \to K$ is uniquely expressible as $\sum_{i=1}^n l_{a_i}\sigma_i$, $a_i \in K$.*

Theorem 31 (Hilbert Theorem 90)**.** *Let K/F be a cyclic Galois extension, with $G = \mathsf{Gal}(K/F) = \langle\sigma\rangle$. Let $0 \neq a \in K$. Then, $\mathsf{Norm}_F^K(\alpha) = 1$ iff there exists $a \in K^\times$ such that $a = \frac{\sigma(\alpha)}{\alpha}$.*

Solvability by Radicals

Definition 32. A Galois extension K/F is {cyclic, abelian, solvable} if $\mathsf{Gal}(K/F)$ is {cyclic, abelian, solvable}.

Definition 33.

1. A *simple radical extension* is $E = F(\alpha)$, where there exists $n \in \mathbb{N}$ such that $\alpha^n \in F$.

2. E/F is a *composite radical extension* if $F \subseteq E_0 \subseteq E_1 = E$ such that each E_{i+1}/E_i is a simple radical extension.

3. A finite extension E/F is a *radical extension* if $F \subseteq E \subseteq E'$, where E' is a composite radical extension.

Theorem 34. *E/F is a radical extension if and only if $E \subseteq K$, K/F is Galois, and $\mathsf{Gal}(K/F)$ is a solvable group.*

Definition 35. A group G is *solvable* if there exists a chain $\{e\} = G_n \trianglelefteq G_{n-1} \trianglelefteq \cdots \trianglelefteq G_0 = G$ such that G_r/G_{r+1} is abelian for all r.

Proposition 36.

1. *Subgroups of solvable groups are solvable.*

2. *Quotients of solvable groups are solvable.*

3. *If G is finite and solvable, there exists a chain with G_i/G_{i+1} cyclic.*

Theorem 37. *If L/F is a solvable extension, then L/F is solvable by radicals.*

Additionally, if L is of the form $F(\alpha_1, \ldots, \alpha_n)$ with $\alpha_i^{n_i} \in F(\alpha_1, \ldots, \alpha_{n-1})$ for all relevant i and $\mathrm{char}(F) = 0$, then there exists a Galois extension $F \subset L \subseteq K$ such that $\mathsf{Gal}(K/F)$ is solvable.

Algebraic Closures and Ring-Theoretic Splitting Fields

Definition 38. A field extension $\bar{F}/F$ is an *algebraic closure* if $\bar{F}$ is algebraic and $\bar{F}$ is algebraically closed.

Theorem 39. *$\mathbb{C}$ is algebraically closed.*

Lemma 40. *If $\bar{F} \supseteq \mathrm{Spl}_F(f)$ for all $f \in F[x]$, then $\bar{F}/F$ is an algebraic closure of F.*

Theorem 41. *Every field F has an algebraic closure.*

Definition 42 (The setup)**.** Let A be a PID with maximal ideal $m = (\pi), \pi$ irreducible. Then, we define F as the fraction field of A and $k = A/(\pi)$ as the residue field.

Theorem 43. *There exists $g \in S_n$ such that $g\mathsf{Gal}(\mathrm{Spl}(\bar{f})/k)g^{-1} \subseteq \mathsf{Gal}(\mathrm{Spl}(f)/F)$.*

Corollary 44. *There exist infinitely many monic polynomials $f \in \mathbb{Z}[T]$ such that $\mathsf{Gal}(\mathrm{Spl}(f)/\mathbb{Q}) = S_n$ for all $n \in \mathbb{N}$.*

Lemma 45. *$R_1 \subseteq R_2 \subseteq R_3$ as subrings of commutative ring R_3. Then, if R_3 is a finitely generated R_2-module and R_2 is a finitely generated R_1-module, then R_3 is a finitely generated R_1-module.*

Lemma 46. *$B = A[\theta_1, \ldots, \theta_n]$ is a finitely generated A-module, with $B^{\mathsf{Gal}(\mathrm{Spl}(f)/F)} = A$.*

Theorem 47. *Consider the map $z(\sigma) : B/m \to B/m$ deduced from the element $\sigma \in \mathsf{Stab}(m)$.*

1. *If $\bar{f} \in k[T]$ is separable, then z is 1-1.*
2. *z is always surjective.*
3. *G acts transitively on $\{m \ni \pi | m$ max ideal of $B\}$, so the groups $\{\mathsf{Stab}(m) \subseteq G\}$ are conjugates of each other.*

Theorem 48. *$G \subseteq \mathsf{Aut}(B)$. Let $A = B^G$, E be a field, and $S = \{\phi : B \to E$ ring homomorphisms$\}$. G acts on S by right multiplication; let $\phi, \psi \in S$. Then, ϕ, ψ belong to the same G-orbit if and only if $\phi|_A = \psi|_A$.*

Theorem 49. *$G \subseteq \mathsf{Aut}(B)$, $A = B^G$. Then,*

1. *Let P be a prime ideal of A. Then, there exists a prime ideal Q of B such that $Q \cap A = P$.*
2. *G acts transitively on $\{Q$ as in 1$\}$.*
3. *$G \hookleftarrow \mathsf{Stab}(Q) \twoheadrightarrow \mathsf{Aut}(\mathsf{Frac}(Q)/\mathsf{Frac}(P))$, where $\mathsf{Frac}(Q)$ is the fraction field of A/P and similarly for Q.*

Valuation Rings and p-adic Numbers

Definition 50. Given objects and morphisms $S_1 \leftarrow S_2 \leftarrow \cdots$, we define the *inverse limit* $\varprojlim S_i := \{(s_1, s_2, \cdots) \,|\, s_i \in S_i, \mathsf{im}(s_{i+1}) = s_i\}$. We can construct the p-adic integers by setting $\mathbb{Z}_p := \varprojlim \mathbb{Z}/p^n\mathbb{Z}$.

Theorem 51. *$f(x) \in \mathbb{Z}[x]$ monic, $\bar{f} \in \mathbb{F}_p[x]$ is the projection of some separable f. Then, $\mathsf{Gal}(\mathrm{Spl}(\bar{f})/\mathbb{F}_p)$ is permutation group-isomorphic to a subgroup of $\mathsf{Gal}(\mathrm{Spl}(f)/\mathbb{Q})$.*

Theorem 52 (Restated from before)**.** *For all $n \in \mathbb{N}$, there exist infinitely many $f(x) \in \mathbb{Z}[x]$ irreducible, degree n, such that $\mathsf{Gal}(\mathrm{Spl}(f)/\mathbb{Q}) \cong S_n$.*

Definition 53. A *normed ring* is a ring R with a norm$||\cdot|| : R \to \mathbf{R}$ such that

1. $||a|| = 0 \iff a = 0$
2. $||ab|| = ||a||\,||b||$
3. $||a+b|| \leq ||a|| + ||b||$.

Definition 54. A *valuation* on a field K is a group homomorphism $v : K^\times \to \mathbb{R}$ such that $v(a+b) \geq \min(v(a), v(b))$. Define $v(0) = \infty$. v is *discrete* if $\mathsf{im}(v) \cong \mathbb{Z} \subseteq \mathbb{R}$. The *valuation ring* of K is the ring $R = \{a \in K \,|\, v(a) \geq 0\}$. R is a *discrete valuation ring* if it is the valuation ring of a discrete valuation.

Proposition 55. *If A is the discrete valuation ring of k, then k is the fraction field of A, and A is a PID.*

Proposition 56. *There exists a norm on k defined by $||a|| = e^{-v(a)}$. More generally, this works for any $0 < \frac{1}{e} < 1$.*

Corollary 57 (Ultrametric Inequality)**.** $||a+b|| \leq \max(||a||\,, ||b||)$.

Proposition 58. *Let v discrete on k, $\hat{k}$ be the completion on k. Then, $\hat{k}$ is a field, and there exists a unique $\hat{v}$ on $\hat{k}$ such that $\hat{v}|_k = v$.*

Definition 59. $\mathbb{Q}_p$ is the completion of $\mathbb{Q}$ with respect to $(\mathbb{Q}, v_p)$, where $v_p\left(p^r \frac{a}{b}\right) = r$.

Proposition 60. $\mathbb{Q}_p = \left(p^{\mathbb{N}}\right)^{-1} \mathbb{Z}_p^\times$, *i.e.* $\mathbb{Q}_p = \{a_n p^n + a_{n+1} p^{n+1} + \cdots | n \in \mathbb{Z}, a_i \in \mathbb{F}_p\}$.

Definition 61. If K is a field with discrete valuation v, define the *residue field* $k = \{a \in K | v(a) \geq 0\}/\{a \in K | v(a) > 0\}$.

Theorem 62 (Hansel's Lemma)**.** *R complete DVR, $f \in R[x]$ monic, $\pi \in R$ such that $v(\pi) = 1$, k residue field, $\bar{f} \in k[x]$ image of f. If $\bar{f} = uv$ with $(u, v) = 1$ and u, v monic, then there exists $g, h \in R[x]$ such that $f = gh, \bar{g} = u, \bar{h} = v$.*

Lemma 63. *K complete field with discrete valuation v_K, L/K degree n extension. Then, there exists a unique v_L which extends v_K, and L is complete with respect to v_L.*

Lemma 64. *Every norm on an n-dimensional vector space over a complete field induces the same topology.*

Theorem 65 (Irreducibility of Cyclotomic Polynomials)**.** *Let* $K = \mathrm{Spl}_{\mathbb{Q}}(x^n - 1)$*. Then,* $\deg(K/\mathbb{Q}) = \varphi(n) = \#(\mathbb{Z}/n\mathbb{Z})^\times$*.*

Theorem 66. *Let* K *be a field with absolute value* $|\cdot|$*. Then, the absolute value extends uniquely to an absolute value defined on the completion* $\hat{K}$*.*

Lemma 67. *Suppose* $a_1, a_2, \ldots \in K$ *with* $a_1 - a_2 \in R\pi, a_2 - a_3 \in R\pi^2$*, etc. Then,* $\lim_{n\to\infty} a_n$ *exists.*

Lemma 68. *Suppose* $G_1, G_2, \ldots \in R[x]$ *such that there exists* $d \in \mathbb{N}$ *such that* $\deg(G_i) \leq d$ *for all* i *and* $G_n \equiv G_{n+1} (\bmod \pi^n)$ *for all* n*, i,e,* $G_n - G_{n+1} = 0$ *in* $(R/\pi^n)[x]$*. Then,* $\lim_{n\to\infty} G_n$ *exists.*

Lemma 69. K *is a field,* $g, h, c \in K[x]$ *with* $(g, h) = 1$*. Then, there exists* $a, b \in K[x]$ *such that* $c = gb + ha$ *with* $\deg(a) < \deg(g)$*.*

Corollary 70 (Corollary 1 of Hansel's Lemma)**.** $f \in R[x]$*. If* $\bar{f} \in k[x]$ *has a root* $\alpha \in k$ *and* $0 \neq f'(\alpha) \in k$*, then there exists* $\beta \in R$ *such that* $f(\beta) = 0$ *and* $\bar{\beta} = \alpha$*.*

Corollary 71 (Corollary 2 of Hansel's Lemma)**.** *Let* $f \in K[x]$*.* $f \in R[x]$ *if and only if* $f(0) \in R$*.*

Lemma 72. *If* $||\cdot||$ *and* $||\cdot||'$ *are norms that satisfy the ultrametric inequality, then they are equivalent to each other.*

Theorem 73. *Let* $||\cdot||$ *be a norm on a finite-dimensional vector space* V *that satisfies the ultrametric inequality, and set* $B = \{v \in V | \, ||v|| \leq 1\}$*. Then,* B *is a finitely generated* R*-module, and* V *is a complete metric space under* $d(x, y) = ||x - y||$*.*

Theorem 74. K *complete with* R, π, k *as usual.* L/K *finite field extension. Define a norm* $||\cdot|| = \left|\mathsf{Norm}_K^L(\cdot)\right|$*.*

1. *This satisfies all required axioms for a norm.*
2. $R_L = \{h \in L | \, ||h|| \leq 1\}$ *is a finitely-generated* R*-module.*
3. R_L *is a ring.*

Lemma 75. *Let* $\alpha \in L$*. Then,* $\mathsf{Norm}_K^L(\alpha) = \left(\mathsf{Norm}_K^{K(\alpha)}(\alpha)\right)^{\deg(L/K(\alpha)}$*.*

Lemma 76. *Assume that the set* $B = \{v \in V | \, ||v|| \leq 1\}$ *is a finitely generated* R*-module. Then,* V *is complete with respect to* $||\cdot||$*.*

Ramification Theory

Definition 77. Given a field extension L/K, define $e(L/K)$ as the *ramification index* and $f(L/K) = \deg(k'/k)$ where $k' = R_L/(\pi')$, where (π') is the unique maximal ideal of R_L. ((π) is the unique maximal ideal of R.)

Proposition 78. $\deg(L/K) = e(L/K)f(L/K)$.

Definition 79. L/K is *unramified* if $e(L/K) = 1$, i.e. $\pi' = \pi$. L/K is *totally ramified* if $f(L/K) = 1$.

Proposition 80. *L/K totally ramified degree n extension, π' generates the maximal ideal of R_L. Then, the monic minimal polynomial of π' is an Eistenstein polynomial $f \in R[x]$, i.e.*

$$f(x) = x^n + a_1 x^{n-1} + \cdots + a_n \tag{4}$$

such that $v(a_i) > 0$ for all i and $v(a_n) = 1$.

Theorem 81. *K algebraically closed field, characteristic 0, $R = k[[z]]$. Then, $K = k((z))$, and there exists a unique field extension L of degree n with $L = K(z^{1/n})$.*

Theorem 82. *Let L/K finite, K complete, k perfect. Then, there exists a unique $K \subseteq K' \subseteq L$ with K'/K unramified and L/K' totally ramified.*

Corollary 83. *If L/K is Galois, then K'/K is Galois.*

Corollary 84. *There exists a natural map $\mathsf{Gal}(K'/K) \xrightarrow{\sim} \mathsf{Gal}(k'/k)$.*

Theorem 85 (Krasnel's Lemma)**.** *Let K be a complete field with a discrete valuation, and let $f = x^n + a_1 x^{n-1} + \cdots + a_n \in K[x]$ with no repeated roots in $\bar{K}$. Then, there exists $\epsilon > 0$ such that for all $g = x^n + b_1 x^{n-1} + \cdots + b_n \in K[x]$ such that $\max_i |a_i - b_i| < \epsilon$, we have $\mathrm{Spl}(f) = \mathrm{Spl}(g)$.*